Revenue Forecaster: a Sequential-Flow Model for Financial Lead Pingtrees

Forecasting revenue and choosing the order of buyers

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Abstract

Lead-generation businesses in consumer finance sell each lead through a *pingtree*: an ordered list of buyers who are offered the lead in sequence until one accepts. We present a forecasting model that turns a client's lead volume into expected monthly revenue by propagating leads through a pingtree built from the buyers' observed acceptance rates and earnings per lead (EPL). We prove that ordering buyers by decreasing EPL maximises expected revenue when acceptances are independent and offers are free, quantify the forecast on anonymised 2026 data (about 115,371 € per 100,000 leads with 20 offers), and show how concentration, diminishing returns, sample-size filters and Monte Carlo uncertainty shape the result. We state clearly where the independence assumption is likely to fail and how the model can be extended.

1 Introduction

A *pingtree* (or waterfall) sells a lead to the first buyer in an ordered list that accepts it. The order matters because every lead accepted early is no longer available to later buyers. A seller that holds a pool of leads therefore needs two answers: how much revenue will the pool generate, and in which order should buyers be called. This paper describes a small forecasting tool, the *Revenue Forecaster*, that answers both from historical acceptance and revenue statistics, and documents the mathematics behind it.

The contribution is deliberately modest. We give (i) a clean sequential-flow model, (ii) a short proof that EPL-descending order is optimal under stated assumptions, (iii) an empirical illustration on anonymised data, and (iv) an honest list of the assumptions that real deployments will violate.

2 Model

Let N be the number of leads entering the pingtree and let buyers be indexed by their position $i = 1, \dots, n$. For buyer i let $a_i \in [0, 1]$ be the *acceptance rate* (accepted leads divided by leads offered) and $e_i \geq 0$ the *earnings per lead* (revenue divided by accepted leads). The number of leads reaching position i is

$$L_i = N \prod_{j<i} (1 - a_j), \tag{1}$$

so buyer i accepts $A_i = a_i L_i$ leads and generates revenue $A_i e_i$. Expected total revenue is

$$R(\pi) = N \sum_{i=1}^{n} a_i e_i \prod_{j<i} (1 - a_j), \tag{2}$$

where π denotes the ordering. The share of leads placed is $1 - \prod_i (1 - a_i)$, which does not depend on the order.

Proposition 1. *Assume acceptances are independent across buyers, an offer to a buyer costs nothing, and a_i, e_i do not depend on position. Then $R(\pi)$ is maximised by ordering buyers by non-increasing e_i.*

Proof. Consider two adjacent buyers at positions k and $k+1$ with parameters (a, e) and (a', e'), and let $\Pi = \prod_{j<k}(1 - a_j)$. Their contribution to R/N in the order $(a, e), (a', e')$ is $\Pi\,[\,ae + (1-a)a'e'\,]$. Swapping them gives $\Pi\,[\,a'e' + (1-a')ae\,]$. The difference is

$$\Pi\,[\,ae + (1-a)a'e' - a'e' - (1-a')ae\,] = \Pi\,aa'(e - e'). \tag{3}$$

The leads that continue past both buyers are the same in either order, so later terms are unchanged. Hence swapping adjacent buyers never lowers revenue when the one with the higher EPL is moved forward. Any ordering can be sorted into EPL-descending order by adjacent swaps, each of which weakly increases R. □

The result is a local index rule in the spirit of sequential search problems [1]. It breaks down when offers are costly, when buyers have monthly caps, or when acceptance depends on position, which we discuss in Section 5. The tool includes a *position penalty* $\rho \in [0, 1]$ that scales the acceptance of the buyer at position i by $1 - \rho\,(i-1)/(n-1)$ to probe the last case.

2.1 Uncertainty

Monthly acceptance and EPL fluctuate. For each buyer we estimate the coefficient of variation of both quantities over the most recent months (bounded between 3% and 80%, with a default of 20% when fewer than two months exist) and draw multiplicative log-normal noise with unit mean. Accepted leads at each position are drawn from a normal approximation to the binomial. We report the 10th, 50th and 90th percentiles of 2,000 simulations.

3 Data and calibration

The reference data are anonymised sources aggregated over 2026: for every buyer, the number of requests, accepted leads, sales and revenue per month. Buyer statistics are pooled over all available 2026 months, so a_i is total accepted over total requests and e_i is total revenue over total accepted. Buyers with no revenue, or with fewer than 50 accepted leads, are excluded; this leaves 40 eligible buyers, of which the top 20 by EPL form the pingtree. Buyers are labelled P1–P20 by position. Some buyers are paid per accepted lead rather than per sale, so their conversion rate is not defined and is shown as unavailable in the tool.

4 Results

4.1 Baseline forecast

For $N = 100{,}000$ leads per month and a 20-offer pingtree, the model forecasts 115,371 € per month, or 1.15 € per lead. 33.0% of leads are placed and the average EPL is 3.50 €. Table 1 lists the pingtree and Figure 4.1 shows how leads are consumed along it.

Table 1: Forecast pingtree for 100,000 leads per month. Acc. is the acceptance rate in %, EPL is in euros and revenue in euros.

Buyer	Leads in	Acc. (%)	Accepted	EPL	Revenue	Share (%)
P1	100,000	0.65	649	29.59	19,202	16.6
P2	99,351	0.65	645	17.31	11,171	9.7
P3	98,706	0.80	787	17.30	13,613	11.8
P4	97,919	1.70	1,666	6.22	10,366	9.0
P5	96,252	0.74	708	4.48	3,170	2.7
P6	95,544	1.00	960	3.88	3,721	3.2
P7	94,585	0.74	702	3.31	2,323	2.0
P8	93,883	0.04	34	3.14	107	0.1
P9	93,849	2.85	2,674	2.71	7,233	6.3
P10	91,175	0.13	114	2.70	308	0.3
P11	91,061	0.46	420	2.35	988	0.9
P12	90,641	2.88	2,613	2.04	5,322	4.6
P13	88,028	0.98	864	2.00	1,728	1.5
P14	87,164	10.38	9,049	1.98	17,933	15.5
P15	78,115	0.78	612	1.84	1,129	1.0
P16	77,503	2.06	1,595	1.77	2,828	2.5
P17	75,908	0.14	106	1.68	177	0.2
P18	75,802	11.01	8,345	1.60	13,346	11.6
P19	67,458	0.50	337	1.60	539	0.5
P20	67,120	0.18	119	1.43	169	0.1

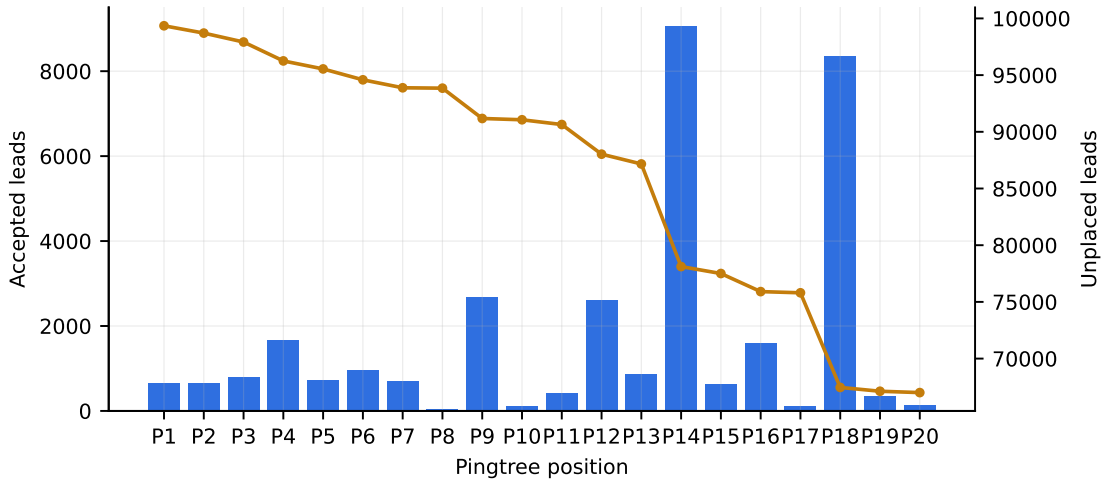


Figure 1: Leads accepted by each buyer (bars) and leads still unplaced after each position (line).

The first buyer sees every lead, but because acceptance rates are low (most below 10%) the pool decays slowly, and about two thirds of leads are still unplaced after the twentieth buyer.

4.2 Concentration and diminishing returns

Revenue is concentrated: the top buyer produces 16.6% of the total and the top three 38.1%; the first 14 buyers reach 80% of revenue (Herfindahl–Hirschman index 1,059). Figure 4.2 shows revenue by position. Adding offers has diminishing returns (Figure 4.2): going from 15 to 20 offers adds 17,059 € per month, and revenue with 10 offers is 38% lower than with 20.

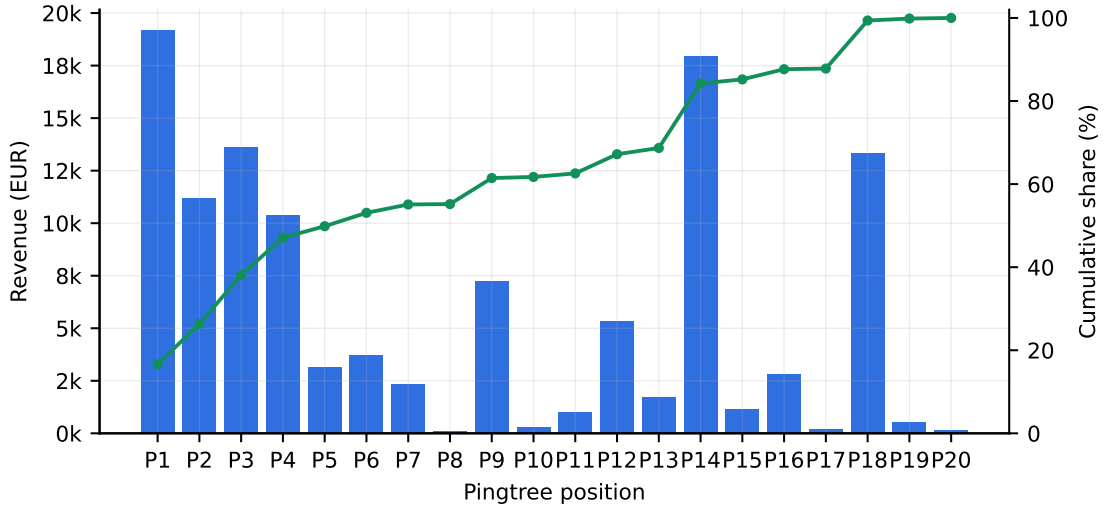


Figure 2: Forecast revenue by position (bars) and cumulative share of the total (line).

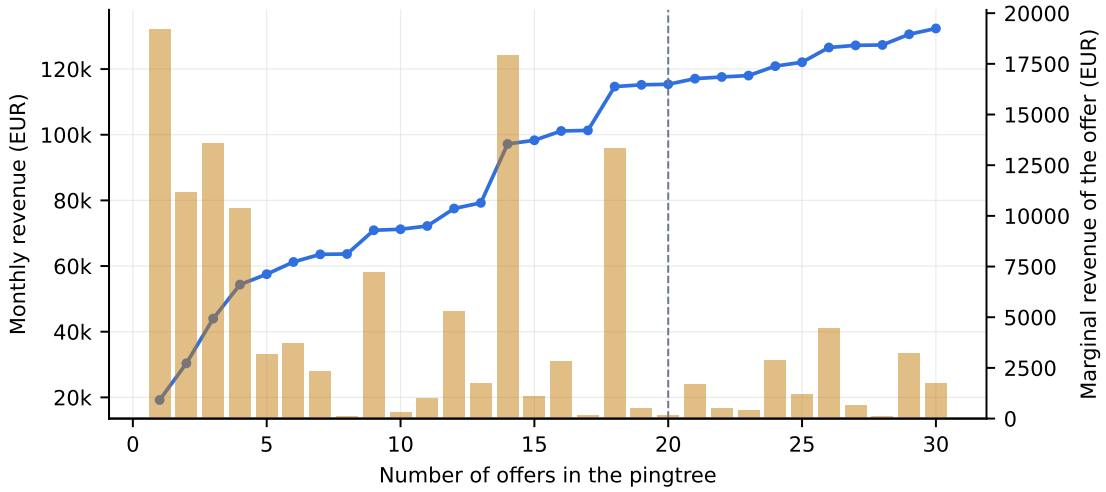


Figure 3: Total revenue as the number of offers grows (line) and marginal revenue of each added offer (bars).

4.3 Does order matter?

Figure 4.3 compares the proposed order with alternatives for the same 20 buyers. Ordering by EPL gives 115,371 €; ordering by acceptance rate gives 98,943 €, by historical revenue 104,577 €, and a random order averages 106,125 €. The worst case, ascending EPL, yields 97,842 €. EPL-descending order is therefore 9% better than a random order, consistent with Proposition 1.

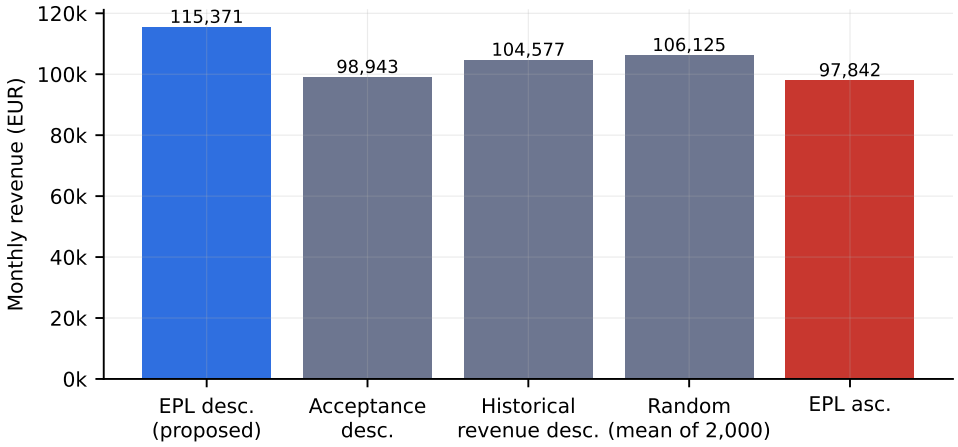


Figure 4: Forecast revenue of the same 20 buyers under different ordering criteria.

4.4 Uncertainty and sensitivity

The Monte Carlo simulation gives a median of 114,135 € with a 10th–90th percentile range of 102,231 € to 129,196 € (Figure 4.4). Figure 4.4 shows that a 10% change in either acceptance or EPL moves the forecast by about 9–10%, a 30% position penalty costs 10%, and cutting the pingtree to 10 offers costs 38%.

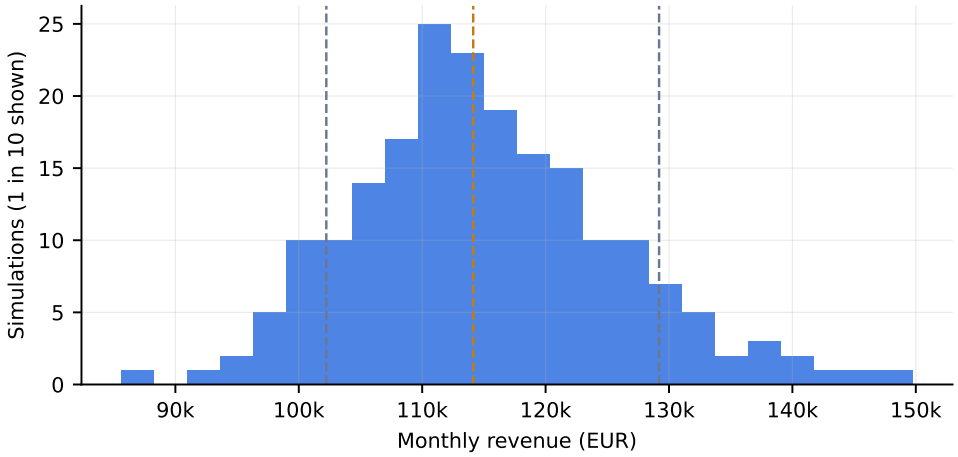


Figure 5: Distribution of monthly revenue over 2,000 simulations; dashed lines mark the 10th, 50th and 90th percentiles.

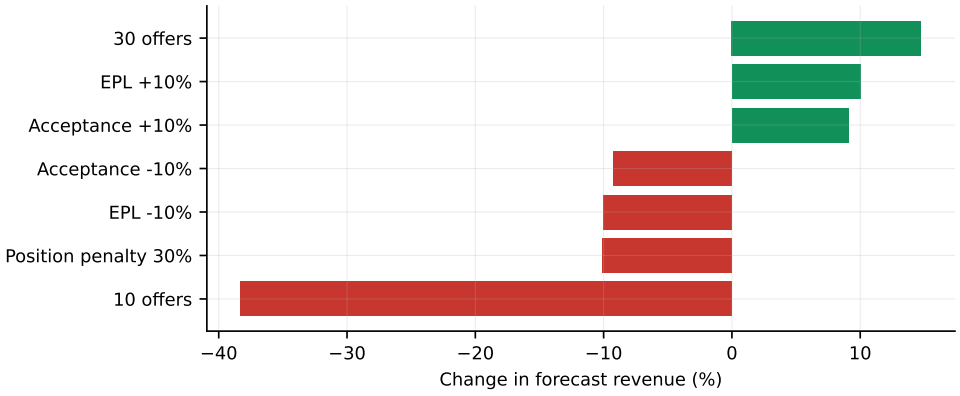


Figure 6: Sensitivity of the forecast to changes in assumptions.

4.5 Sample-size filter and month-to-month variation

Small samples are dangerous. Without any minimum on accepted leads the forecast rises to 4,500,000 €, because a handful of buyers with very few accepted leads show an implausibly high EPL and move to the top of the tree. With a minimum of 50, 200 and 1,000 accepted leads the forecasts are 115,371 €, 91,388 € and 100,210 €. The default of 50 is a compromise, not a recommendation: the right threshold should be chosen by cross-validating forecasts against realised revenue. Forecasts based on January alone and February alone are 107,246 € and 90,022 €, a spread that the Monte Carlo range only partly captures and that is a reminder that two months of data are a thin base.

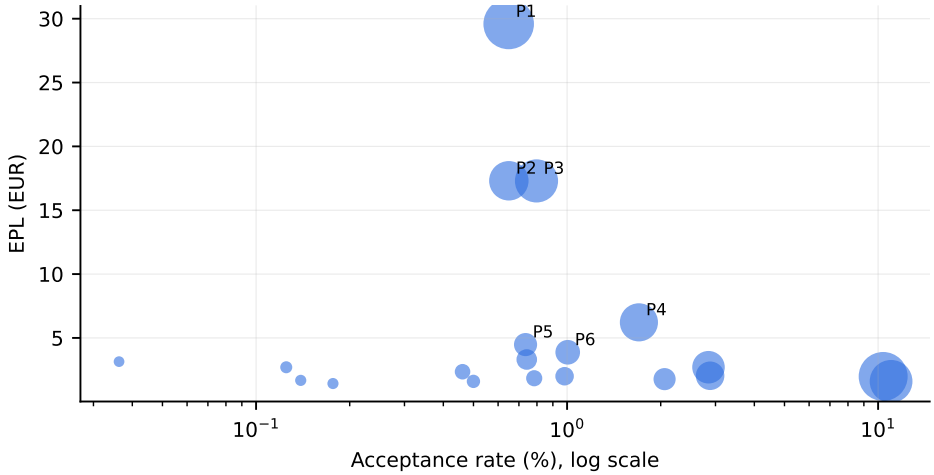


Figure 7: EPL against acceptance rate for the 20 buyers. Bubble area is forecast revenue; the horizontal axis is logarithmic.

5 Limitations and extensions

Position-dependent acceptance. Acceptance rates were measured in the positions buyers actually occupied. Buyers placed lower see leads already rejected by others, which may have lower quality, so their acceptance in a new tree could be smaller. The position-penalty parameter lets users test this, but the correct approach is to estimate acceptance by position from experiments.

Caps and costs. Many buyers impose monthly caps. With caps the optimal order is no longer an index rule and a small optimisation problem must be solved. If pings have a cost, buyers with low acceptance and moderate EPL may not be worth calling.

Lead types. The quality adjustments for lead types in the tool are editable hypotheses. They are not estimated from data and should be replaced by segment-level statistics.

Independence. Acceptance decisions are modelled as independent. In practice buyers use similar eligibility rules, so a lead rejected by one is more likely to be rejected by another, which would make the forecast optimistic.

Forecast, not guarantee. The output is an expected value under stated assumptions, not a commitment.

6 Conclusion

A sequential-flow model with a simple ordering theorem is enough to build a transparent revenue forecaster for lead pingtrees. On anonymised 2026 data the forecast for 100,000 leads is about 115,371 € per month with a plausible range of 102,231 € to 129,196 €. The main practical risks are small samples, position effects and caps; each can be addressed with the extensions above.

References

[1] M. L. Weitzman. Optimal search for the best alternative. *Econometrica*, 47(3):641–654, 1979.

[2] P. Glasserman. *Monte Carlo Methods in Financial Engineering*. Springer, 2004.